Aryan Verma

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GitHub - LinkedIn - Portfolio

Professional Summary

Software developer with an M.Sc. in Advanced Computer Science (University of Birmingham) and hands-on experience building web applications and APIs with React.js, TypeScript, Node.js, Java/Spring Boot and Python/Django, backed by PostgreSQL and AWS. Gained across four technical internships, including frontend development for an aviation booking platform and an applied machine-learning project, plus personal projects built end to end. Works with Git, testing and Agile practices. Seeking a graduate or junior software developer role.

Education

M.Sc. Advanced Computer Science *Sep 2025 - Oct 2026*

University of Birmingham, Birmingham, UK

- Grade: Merit
- Modules: Computer Vision, Human-Computer Interaction, Secure Systems, Intelligent Interactive Systems, Intelligent Software Engineering

Semester Exchange Programme *Mar 2025 - Jun 2025*

VSB - Technical University of Ostrava, Ostrava, Czech Republic

- GPA: 4.0/4.0

B.Tech (Hons.) Computer Science & Engineering (IoT & IS) *Aug 2021 - Jul 2025*

Manipal University Jaipur, Jaipur, India

- CGPA: 8.19 - Dean's List Excellency Award (7th Semester)
- Modules: Data Structures & Algorithms, Operating Systems, Database Systems, Software Engineering

Professional Experience

Programmer Analyst Trainee *Jul 2025 - Sep 2025*

Cognizant Technology Solutions, Pune, India, On-site

- Strengthened proficiency in Java, Spring Boot and React through instructor-led training and assessed coursework within a professional SDLC environment.
- Implemented features for an early-stage e-commerce platform using React and Spring Boot under manager supervision before relocating to the UK for postgraduate study.

Frontend Developer *Jan 2025 - May 2025*

ASR Aviation, Jaipur, India, Remote

- Delivered the frontend of an aviation booking platform supporting high-volume concurrent user sessions.
- Built 5+ core frontend modules, including flight booking, operator listings and administration dashboards, using reusable components to streamline internal development.
- Established secure, scalable AWS backend infrastructure and contributed to Agile testing, debugging and release cycles.

Python/Django Intern *Jul 2024 - Dec 2024*

Zummit Infolabs, Bengaluru, India, Remote

- Completed backend development training in Python/Django, Node.js, Express.js and RESTful API design.
- Gained exposure to microservices architecture, database integration and API testing through structured exercises and code reviews.

Data Science Intern *May 2024 - Jul 2024*

Celebal Technologies, Jaipur, India, Remote

- Developed a spam email detection system using data preprocessing, feature engineering and model training across Naive Bayes, SVM, Random Forest, Gradient Boosting and CNN.
- Achieved 92% classification accuracy against an 82% target benchmark through systematic model evaluation and selection.
- Deployed the final solution as a functional Python pipeline using scikit-learn and pandas.

Projects

CodeClassroom - dpa.shah.fyi *React, TypeScript, CSS, Spring Boot, JHipster, JDL*

- Created a full-stack visual domain modelling platform with a custom compiler on top of JHipster, extending its domain language to support object-oriented programming patterns beyond native JDL capabilities.
- Designed a visual modelling environment that simplifies design-pattern creation and full application generation by compiling visual models into application-ready code, with both frontend and backend deployed live.

Optimised Clinic Appointment System *React.js, Node.js, PostgreSQL, AWS*

- Streamlined scheduling logic with robust edge-case handling to reduce booking errors and manage concurrent session reliability on AWS infrastructure.

Savings Goal Tracker *React.js, Chart.js, Tailwind CSS*

- Secured 3rd place at IEEE challenge; built interactive goal tracking and visual analytics using Chart.js for real-time visualisation.

Self-Driving Car Prototype *Arduino, C/C++, Ultrasonic Sensors*

- Engineered a miniature autonomous vehicle with real-time obstacle detection and avoidance using ultrasonic sensors and C++ firmware for navigation and steering control.
- Applied hardware-software integration and low-level programming principles across embedded systems.

Technical Skills

- **Programming Languages:** Python, Java, JavaScript, TypeScript
- **Frameworks & Libraries:** Spring Boot, Node.js, React.js
- **Web:** HTML, CSS
- **Core Concepts:** REST APIs, Microservices, SQL & NoSQL Databases, Debugging, System Integration, CI/CD, Agile/SDLC
- **Systems & Tools:** Git, Postman, Selenium, Docker
- **Cloud & Databases:** AWS (EC2, S3), MySQL, MongoDB
- **Embedded / IoT:** Arduino, Ultrasonic Sensors, RF Modules, Bluetooth, C++ Firmware

Certifications

- Foundations of Software Testing and Validation - Coursera
- Selenium WebDriver with Java - Coursera
- The Complete Oracle SQL Bootcamp (2024) - Udemy
- HackerRank SQL - 4-Star Badge
- Web Developer Bootcamp 2024 - Udemy
- Python Bootcamp - Udemy

Entry-level graduate, open to UK graduate salary levels. Applying for the UK Graduate visa (2 years' right to work, no sponsorship needed during this period).